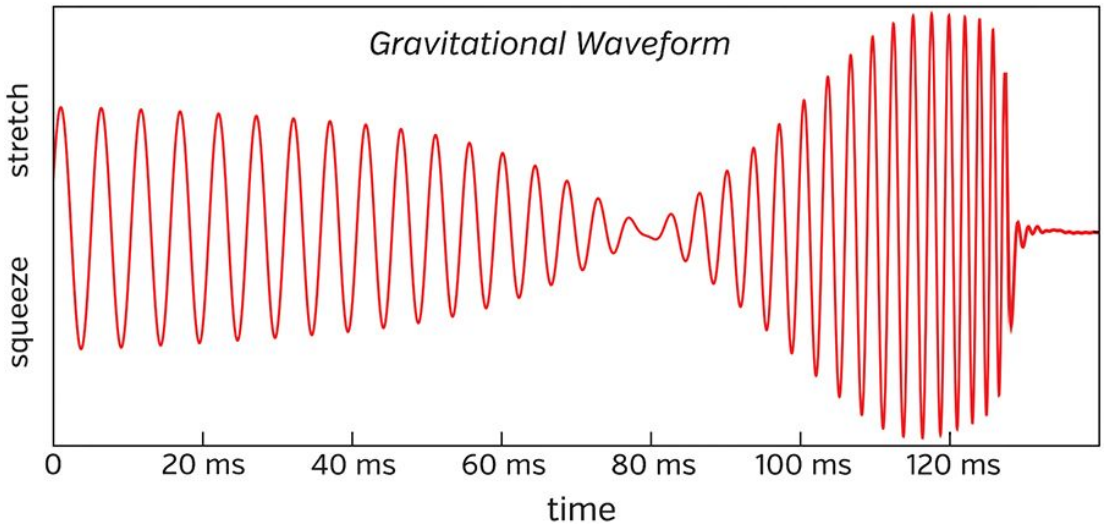


Fig. 16.2. The last 120 milliseconds (ms) of the gravitational waveform discovered by LIGO. *[Drawing by Kip based on simulations by Yanbei Chen and by Foucart et al. (2011).]*

By comparing the waves' shape (their "waveform"; Figure 16.2) with simulations performed on supercomputers, Professor Brand and his team deduced their source.

Neutron Star Orbiting a Black Hole

A neutron star, orbiting around a black hole, had emitted the waves. The star weighed 1.5 times as much as the Sun, the hole weighed 4.5 times the Sun, and the hole was spinning rapidly. The spin dragged space into motion, and the space whirl grabbed the star's orbit, forcing it to precess slowly, like a tilted top. The precession modulated the waves, causing them to rise and fall in amplitude (Figure 16.2).

The waves traveled out through the universe, carrying away energy (Figure 16.3). With its energy gradually decreasing, the star gradually spiraled inward toward the black hole. When the distance between the star and the hole had shrunk to 30 kilometers, the hole's tidal gravity began tearing the star apart. Ninety-seven percent of the stellar debris was swallowed by the black hole, and 3 percent was thrown outward,